Eigen-CNN: Eigenimages Plus Eigennoise Level Maps Guided Network for Hyperspectral Image Denoising

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***Abstract*— In recent years, neural network-based methods have shown promising results in hyperspectral image (HSI) denoising areas. Real HSIs exhibit the substantial variations in noise distribution due to various factors, such as different imaging techniques, camera variations, imaging environments, and hardware aging. In this article, we develop an eigenimage plus eigennoise level map-guided convolutional neural network (CNN) for HSI denoising. Our main idea is to perform eigendecomposition on HSIs, utilize the low-rank property of HSIs in the spectral dimension and approximate the spectral vectors in a low-dimensional orthogonal subspace, where representation coefficients are called eigenimages. Besides eigenimages, we make use of estimated eigennoise level map as an input to guide the network for denoising. The proposed network can be constructed without restriction in the number of eigencomponents by using all eigenimages and eigennoise level maps of training noisy-clean pairs. In the inference part, the trained network can be used to remove noise in observed eigenimages without restriction in the number of eigencomponents, and an underlying clean image HSI can be estimated by performing orthogonal projection back. Experimental results on both simulated and real HSIs demonstrate the effectiveness of our trained Eigen-CNN compared with state-of-the-art HSI denoising methods. A MATLAB demo of this work is available at: https://github.com/LinaZhuang/HSI-denoiser-Eigen-CNN for the sake of reproducibility.**

***Index Terms*— Convolutional neural network (CNN), deep neural network, hyperspectral image (HSI) restoration, low-rank representation, subspace representation.**

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NOMENCLATURE

$\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$	3-D tensor (calligraphic letter).
$\mathbf{X}$, $\mathbf{x}$, and x	Matrix (boldface capital letter), vector (boldface lowercase letter), and scalar (italic lowercase letter).
mode n	nth dimension of a tensor.
$\mathbf{X}_{(3)} \in \mathbb{R}^{I_3 \times (I_1 * I_2)}$	Mode-3 unfolding of $\mathcal{X}$. A tensor can be unfolded into a matrix by rearranging its mode-3 vectors, which are the column vectors of $\mathbf{X}_{(3)}$.
$\mathcal{X} \times_3 \mathbf{E}$	Tensor matrix multiplication. Mode-3 product of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ by a matrix $\mathbf{E} \in \mathbb{R}^{J_n \times I_3}$ is a tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times I_2 \times J_n}$, denoted by $\mathcal{Y} = \mathcal{X} \times_3 \mathbf{E}$, which is corresponding to a matrix multiplication, $\mathbf{Y}_{(3)} = \mathbf{E}\mathbf{X}_{(3)}$.
$\|\mathcal{X}\|_F$	Definition of Frobenius norm of a matrix is extended to a tensor as follows: $\|\mathcal{X}\|_F = \sqrt{\sum_{i_1,i_2,i_3} \|x_{i_1,i_2,i_3}\|^2}$.

I. INTRODUCTION

HYPERSPECTRAL imaging sensors capture electromagnetic energy in their instantaneous field of view across hundreds or even thousands of spectral channels, boasting superior spectral resolution compared with multispectral or RGB cameras [1], [2], [3], [4], [5], [6], [7]. This enhanced spectral resolution allows for precise identification of materials and objects within the images [8]. Such sensors have paved the way for numerous novel applications centered around material identification, including precision agriculture, target detection [9], [10], [11], [12], [13], and environmental conservation.

Hyperspectral imaging sensors offer a high spectral resolution, enabling the capture of detailed information about the electromagnetic spectrum. However, this advantage comes at the cost of reduced signal energy per spectral channel, leading to a higher level of noise compared with traditional multispectral or RGB cameras. The presence of measurement noise poses a significant challenge to the widespread utilization of hyperspectral images (HSIs) in various applications that require precise material identification and analysis [14]. Therefore, denoising HSIs have emerged as a critical inverse

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problem in the field of HSI processing [1], [15], [16], [17]. By effectively reducing the noise intensity, denoising techniques play a crucial role in enhancing the quality and reliability of hyperspectral data, facilitating accurate material identification, and improving the overall performance of hyperspectral imaging systems.

Cutting-edge HSI denoising techniques can be primarily classified into two categories: those based on traditional machine learning and those employing deep learning approaches. Traditional machine learning-based methods utilize proper tools to characterize the data structure of HSIs. For instance, the high spatial–spectral correlation enables an HSI to be approximated by a low-rank matrix/tensor [18], [19], [20], [21], [22], [23]. Also, high-dimensional noise-free spectral vectors can be represented sparsely in an over-complete dictionary (total variation-regularized low-rank matrix factorization (LRTV) method [24]), total variation domain (spatio-spectral total variation (SSTV) method [25]), wavelet transform domain [16], [26], and orthogonal subspace domain [27]. The core principle behind most Gaussian denoisers is that Gaussian noise is random and densely distributed in the aforementioned domains, allowing for effective separation of the image signal from Gaussian noise. In HSIs, stripe noise, deadlines, and impulse noise are generally represented as a sparse component, such as ℓ_1 norm in [19] and [28], $\ell_{2,1}$ norm in [29], and nonconvex sparsity norm in [30]. Despite their high accuracy, these traditional machine learning-based techniques often involve solving intricate regularized optimization problems iteratively, resulting in a lengthy process and unstable outcomes that are highly dependent on the configuration of regularization parameters. A parameter-free HSI noise removal method, FastHyMix, is introduced in [31], which characterizes the complex distribution of mixed noise by using a Gaussian mixture model [32].

Among traditional machine learning-based HSI denoising methods, the spectral low-rank representation methods underlie the state-of-the-art performance. The spectral domain exhibits a higher correlation than the spatial domain, prompting researchers to take advantage of this signal redundancy to address hyperspectral imaging inverse problems, such as denoising [31], inpainting [18], superresolution [33], [34], spectral unmixing [35], hyperspectral and multispectral image fusion [34], [36], [37], hyperspectral anomaly detection [14], [38], and compressive sensing [39], [40]. There are several ways to enhance the low-rankness of spectral vectors in an HSI, including imposing a nuclear norm on the corresponding image matrix [30], factorizing the image matrix into two smaller matrices [19], and representing high-dimensional vectors in a low-dimensional orthogonal subspace [18], [31]. This article follows the research line of orthogonal subspace representation of spectral vectors and refers to the coefficients of the subspace representation of an HSI as *eigenimages*.

Investigating the data structure of eigenimages has led to the development of sophisticated HSI denoising methods, which can be classified into two groups based on whether they utilize traditional machine learning or deep learning techniques for further eigenimage estimation. The former group typically employs various nonlocal patch-based or total variation-based regularizations to characterize eigenimages. For example, the FastHyDe method [18] utilizes BM3D to remove residual noise in eigenimages. The global local factorization (GLF) method [27] exploits the self-similarity of eigenimages through a low-rank tensor factorization of nonlocal similar 3-D patches. The NG-meet method [41] employs matrix-based weighted nuclear norm minimization (WNNM) to denoise each nonlocal patch group of eigenimages. The double-factor regularized low-rank tensor factorization (LRTF-DFR) method [19] minimizes the total variation in the spatial dimension of eigenimages. The latter group estimates eigenimages by incorporating neural networks into denoising optimization problems. For instance, the HySuDeep method [42] uses a single-band CNN to filter each eigenimage, while the more recent SLRP-DSP method [43] employs a self-supervised network to learn the deep spatial prior of each eigenimage.

In recent years, deep learning-based HSI denoising has emerged as a hot research topic [44], [45], [46], [47], [48], [49], [50], [51], [52], [53], [54], [55], [56], attracting increasing attention from the remote sensing community. Several notable research achievements have contributed to the advancement of this field. One of the earlier works in this area is the HSI-DeNet method proposed by Chang et al. [57]. HSI-DeNet utilizes a convolutional neural network (CNN) and incorporates techniques, such as residual learning, dilated convolution, and multichannel filtering to effectively denoise HSI data. Yuan et al. [58] introduced the spectral–spatial deep CNN method (HSID-CNN), which takes as input a single spectral band along with its neighboring bands. The target of the network is to remove the noise from the central band. This design allows the trained network to be applied to new images with different spectral bands, but it sacrifices the utilization of all spectral bands during the denoising process. Similar to HSID-CNN, the HSI single denoising CNN method (HSI-SDeCNN) [59] also takes a single band as input along with its neighboring bands. However, HSI-SDeCNN improves the efficiency of the denoiser by introducing a reversible downsampling operator, which reshapes the input image into four downsampled subimages. Another representative work in this field is the alternating directional 3-D quasi-recurrent neural network for HSI denoising (QRNN3D) [60]. QRNN3D utilizes 3-D convolution to simultaneously extract both spatial and spectral features. It also incorporates a quasi-recurrent pooling unit to capture spectral information.

In contrast to traditional machine learning-based approaches for HSI denoising, deep learning-based methods utilize a vast amount of noisy-clean paired training data to learn nonlinear denoising functions. This approach eliminates the need for designing complex hand-crafted priors and solving computationally expensive optimization problems. Although deep learning-based denoising methods have shown promising results in research, one of the challenges is network generalization. That is, when a well-trained network is used on new HSI with significantly different data distributions (for example, the number of image bands) than the training data, the denoising performance of the network can be poor. Furthermore, due to the complex and diverse imaging objects (i.e., surface objects) in remote sensing images, different sensor noise distributions,

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and the changes in the noise distribution caused by hardware aging of the same sensor, it is not feasible to obtain a sufficiently diverse dataset for network training in practice.

A. Contributions

The motivation of our study is to address the above mentioned generalization issue in deep HSI denoising methods. To avoid limiting the number of image bands that the network can handle by using the original HSI as input, we combine low-rank representation and neural network techniques to construct a new HSI denoising framework, called Eigen-CNN. An HSI has high spectral correlation, which allows their spectral vectors to be approximately represented in a low-dimensional orthogonal subspace. We call the representation coefficients eigenimages, which have high spatial correlation and can be efficiently denoised using CNN techniques. By utilizing eigenimages, our method can effectively denoise HSIs without being limited by the number of image bands. Furthermore, by investigating the data structure and noise distribution of eigenimages, we propose eigenimages plus eigennoise level maps guided neural network for HSI denoising. Contributions of this work are summarized as follows.

1) This article proposes a novel HSI denoising framework, called Eigen-CNN, which utilizes the low-rank property of spectral dimension through eigendecomposition of the observed data spectral correlation matrix and exploits the spatial correlation of images through CNN to achieve high-precision denoising.
2) Compared with other deep HSI denoising networks that denoise in the original image domain, our method works on eigenimages, and our network is eigenimage-wise, which helps to address one of the challenges of deep HSI denoising. Our trained Eigen-CNN can be well applied to new HSIs without limitation of the number of image bands.
3) Eigendecomposition has been widely utilized in traditional machine learning-based HSI denoisers. However, in contrast to these approaches, Eigen-CNN leverages a deep network for eigenimage denoising, avoiding the need for designing complex hand-crafted priors and solving computationally expensive optimization problems. Furthermore, Eigen-CNN improves the denoising performance by designing a special network input, i.e., leveraging a high-quality first eigenimage and a noise level map of eigenimage (referred to as eigennoise).
4) The use of eigennoise level maps as one of the network inputs enables our network to effectively handle spatial-variant noise. A challenge in integrating noise level maps into the input data lies in estimating the noise in the eigendomain of HSIs, and this article presents two methods for estimating eigennoise level maps.

The outline of this article is as follows. We formulate the problem and investigate the data structure of eigenimages in Section II. The Eigen-CNN framework is then introduced in Section III. Sections IV and V present numerical examples that demonstrate the performance of the proposed approach for synthetic and real datasets, respectively. Finally, concluding remarks are given in Section VI.

II. Related Works

We remark that the Eigen-CNN proposed in this article is connected to our previous denoising works [14], [18], [27], [31], [32], [61], where the core framework of these methods all involves eigendecomposition to perform denoising on eigenimages rather than in the original image space. However, the main difference among these methods lies in how to remove noise from eigenimages under different types of noise, such as mixed noise (i.e., a mixture of Gaussian noise, stripe noise, and impulse noise) and pure Gaussian noise. Research focusing on mixed noise includes [31], [32], for example, FastHyMix [31] investigates how to estimate high-precision subspace $\mathbf{E}$ under mixed noise conditions, aiming to predominantly retain easily processable Gaussian noise on the eigenimages, as opposed to challenging mixed noise. The research emphasis lies in the estimation of the signal subspace under mixed noise conditions and the noise probability distribution of eigenimages.

On the other hand, our current work falls into the second category, specifically under the assumption of pure Gaussian noise. Our related previous works include [14], [18], [27], [61], which concentrate on the more efficient removal of noise from eigenimages under Gaussian noise conditions, aiming to improve the speed or accuracy of denoising. The emphasis of the research lies in the design of denoisers for eigenimages. For example, the FastHyDe method [18] exploits the self-similarity characteristics of eigenimages in the spatial domain to denoise each eigenimage individually; the GLF method [27] utilizes the correlation between eigenimages and their low rankness in the spatial dimension, denoising all eigenimages together through low-rank tensor approximation techniques; the RhyDe method [14] considers denoising in the presence of anomalous targets, with the goal of preserving the signal of anomalous targets in the denoised image; the E2E method [61] is based on self-supervised learning technique, training the eigenimage denoising network using only noisy images (rather than paired noisy-clean image pairs). These methods essentially remove noise from eigenimages using different techniques, each with its own advantages and disadvantages. For example, FastHyDe is fast but relatively less accurate in denoising; GLF is slow but relatively more accurate in denoising; RhyDe addresses the potential issue of losing the signal of anomalous targets in the results of FastHyDe and GLF, but introduces a new variable representing the signal of anomalous targets into the model, making the model computation more complex and requiring manual adjustment of regularization parameters; E2E provides a solution to the problem of lack of clean training images in the eigenimage denoising network by using a self-supervised learning technique, but inevitably introduces errors in pseudolabel generation, leaving the possibility for further improvement in network accuracy after training. The proposed method, Eigen-CNN, in this article belongs to the framework of eigenimage denoising but differs in how to remove residual noise from eigenimages. Compared with traditional machine learning-based methods (such as FastHyDe, GLF, and RhyDe), Eigen-CNN uses a

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deep network for eigenimage denoising, avoiding the need for designing complex hand-crafted priors and solving computationally expensive optimization problems. Our experiments in this article demonstrate that Eigen-CNN further improves denoising accuracy. In comparison to the E2E method, Eigen-CNN uses eigennoise level maps to guide the network to further remove residual noise from eigenimages. Moreover, this article investigates how to estimate noise in the eigendomain [see (11) and (12)], and the use of eigennoise level maps enables our Eigen-CNN to handle spatially variant noise.

III. Problem Formulation

A. Observation Model

Some notations and algebraic operations used in this article and their definitions are provided in the Nomenclature. Let $\mathcal{X} \in \mathbb{R}^{r\times c\times b}$ be an underlying clean HSI with t ($t = r \times c$) pixels and b bands. Assuming that noise is additive, we can write an observation model as follows:

$$\mathcal{Y} = \mathcal{X} + \mathcal{O} \tag{1}$$

where $\{\mathcal{Y}, \mathcal{O}\} \in \mathbb{R}^{r\times c\times b}$ denote an observed HSI and Gaussian noise, respectively.

B. Eigenimages of an HSI

Due to the extremely high spectral correlation, hyperspectral vectors can be represented well in a low-dimensional subspace (see Fig. 1) [1], [14], [27], i.e.,

$$\mathcal{X} = \mathcal{Z}\times_3\mathbf{E} \tag{2}$$

with $\mathbf{E} = [\mathbf{e}_1, \cdots, \mathbf{e}_p] \in \mathbb{R}^{b\times p}$ ($p \ll b$) and $\mathcal{Z} \in \mathbb{R}^{r\times c\times p}$. Columns of $\mathbf{E}$ hold an orthogonal basis for the signal subspace, then we have $\mathbf{E}^T\mathbf{E} = \mathbf{I}_p$ with $\mathbf{I}_p$ representing the identity matrix of size p-by-p. The entries of $\mathcal{Z}$ are representation coefficients of $\mathcal{X}$ with respect to $\mathbf{E}$. Hereafter, mode-3 slices of $\mathcal{Z}$ are termed the *eigenimages*. To simplify the analysis, let $\mathbf{Z}_i \in \mathbb{R}^{r\times c}$ be ith eigenimage $\mathcal{Z}(:, :, i)$.

As long as the noise is Gaussian distributed, spectral subspace, $\mathbf{E}$, can be learned with high precision from the noisy HSI (see a detailed explanation in [14, Sec. II-B1]. In this article, we obtain a spectral subspace by simply implementing eigendecomposition of the observed data spectral correlation matrix, denoted by $\mathbf{R}_y = \mathbf{Y}_{(3)}\mathbf{Y}_{(3)}^T/t$. Then, we have $\mathbf{R}_y = \mathbf{U}\Sigma^2\mathbf{U}^T$, where $\mathbf{U} \in \mathbb{R}^{b\times b}$ and $\Sigma^2 \in \mathbb{R}^{b\times b}$ are the eigenvector and eigenvalue matrices of the correlation matrix $\mathbf{R}_y$, respectively. The eigenvalues in diagonal matrix Σ^2 are ordered by decreasing magnitude. The spectral subspace is approximately spanned by p eigenvectors associated with the p largest eigenvalues, i.e.,

$$\mathbf{E} = \mathbf{U}(:, 1 : p). \tag{3}$$

The dimension of subspace, p, can be automatically estimated by the HySime method [62] and virtual dimensionality (VD) method [63].

Given $\mathcal{Y}$ and an estimate of $\mathbf{E}$, we can generate noisy eigenimages by projecting the noisy HSI $\mathcal{Y}$ onto the prelearned orthogonal subspace

$$\widetilde{\mathcal{Z}} = \mathcal{Y}\times_3\mathbf{E}^T \tag{4}$$

where $\widetilde{\mathcal{Z}} \in \mathbb{R}^{r\times c\times p}$ is the noisy eigenimages, which are corrupted by Gaussian noise.

Consequently, the HSI denoising problem can be turned into an estimation problem of eigenimages given noisy eigenimages $\widetilde{\mathcal{Z}}$. Unlike most CNN-based HSI denoising methods, the eigenimage estimation scheme enables us to denoise HSIs with the arbitrary number of bands.

In Sections III-C and III-D, we discuss the data structure of eigenimages and the distribution of noise remaining in eigenimages.

C. Guidance of First Eigenimage

This article explores a useful characteristic of eigenimage data; that is, the first eigenimage learned from a noisy HSI has a much higher signal-to-noise ratio (SNR) than the remaining eigenimages, which inspires us to take the high-quality first eigenimage to guide the restoration of remaining eigenimages.

Claim 1: For a given HSI corrupted by Gaussian i.i.d. noise, the initial eigenimage (associated with the maximum eigenvalue) exhibits a superior SNR compared to the remaining eigenimages.

Justification: For a distribution with Gaussian i.i.d. noise, the correlation matrix of the noise can be represented as $\mathbf{R}_n = \nu^2\mathbf{I}_b \in \mathbb{R}^{b\times b}$, where ν^2 represents the variance of the noise per band. By utilizing the eigendecomposition of the correlation matrix of the signal, we obtain $\mathbf{R}_x = \mathbf{X}_{(3)}\mathbf{X}_{(3)}^T/t = \mathbf{U}\Sigma_{signal}^2\mathbf{U}^T$, where $\mathbf{U} = [\mathbf{u}_1, \mathbf{u}_2, \cdots] \in \mathbb{R}^{b\times b}$ and $\Sigma_{signal}^2 = \mathrm{diag}(\sigma_1^2, \sigma_2^2, \cdots) \in \mathbb{R}^{b\times b}$ denote the eigenvector and eigenvalue matrices of $\mathbf{R}_x$, respectively. The eigenvalues in the diagonal matrix Σ_{signal} are arranged in descending order of magnitude. The SNR of the ith eigenimage can be computed as follows:

$$\mathrm{SNR}_i = 10\log_{10}\frac{\mathbf{e}_i^T\mathbf{R}_x\mathbf{e}_i}{\mathbf{e}_i^T\mathbf{R}_n\mathbf{e}_i} = 10\log_{10}\frac{\sigma_i^2}{\nu^2}. \tag{5}$$

As we arrange the eigenvalues σ_i^2 ($i = 1, 2, \cdots, p$) in descending order of magnitude, the SNR of the ith eigenimage is higher than that of the $(i + 1)$th eigenimage. In addition, in practical scenarios, the value of σ_1^2 is typically much larger than the noise variance, resulting in a higher SNR for the first eigenimage.

In reality, noise in real HSIs is not perfectly i.i.d. However, considering the fact that the image signal is much stronger than the noise signal in observed HSIs, i.e., $\sigma_1^2 \gg \nu^2$, the first eigenimage (corresponding to the largest eigenvalue) is usually high quality and can be used to guide the restoration of remaining eigenimages.

D. Eigennoise Level Map

The estimation of noise in HSIs is an active and fundamental research area, with numerous outstanding algorithms already established, such as HySime [62] and the homogeneous regions division and spectral decorrelation (HRDSDC)

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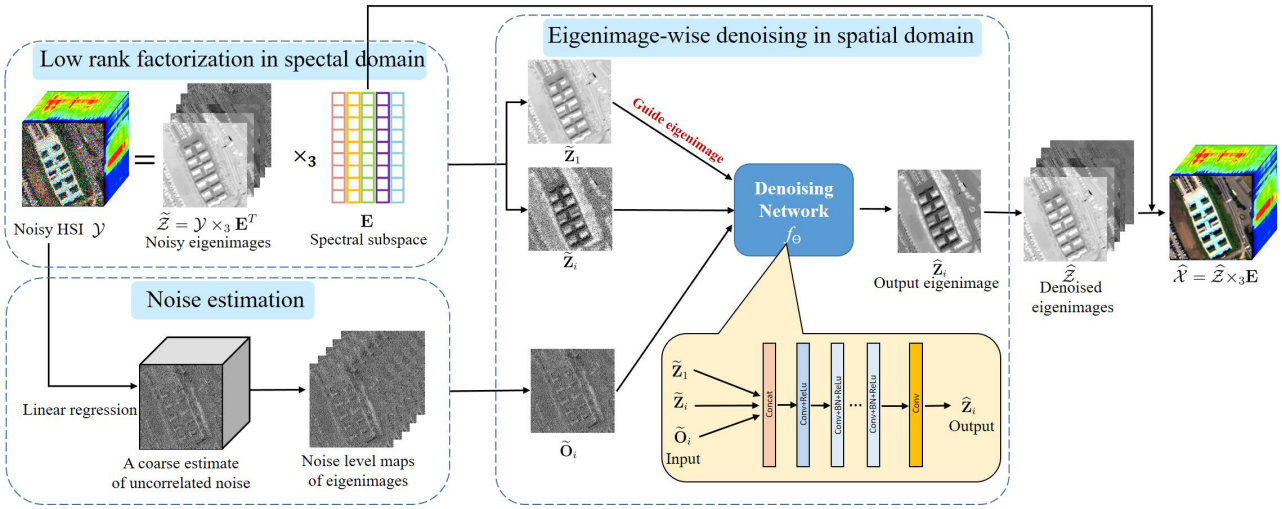


Fig. 1. Details of the proposed eigenimages plus eigennoise level maps guided network.

method [64]. Here, we exploit spectral correlation of HSIs, and simply implement a linear regression method[1] to estimate a coarse noise (see more details in [31]).

We assume that one band can be approximately represented as a linear combination of the remaining $(b-1)$ bands [62], [65], i.e.,

$$\left[\mathbf{Y}_{(3)}^T\right]_{:,j} = \left[\mathbf{Y}_{(3)}^T\right]_{\partial_j} \boldsymbol{\beta}_j + \boldsymbol{\xi}_j \tag{6}$$

where $\mathbf{Y}_{(3)}^T$ denotes the transpose of the mode-3 unfolding matrix $\mathbf{Y}_{(3)}$, the subscript $[\cdot]_{:,j}$ means extracting jth column from a matrix, a matrix with the subscript $[\cdot]_{\partial_j}$ means the matrix including all columns except jth column, $\boldsymbol{\beta}_j \in \mathbb{R}^{b-1}$ denotes the regression coefficients, and $\boldsymbol{\xi}_j \in \mathbb{R}^I$ (with $I = r \times c$) denotes the regression error. The regression coefficients $\boldsymbol{\beta}_j$ can be estimated by the least squares method, i.e.,

$$\begin{aligned}\hat{\boldsymbol{\beta}}_j &= \arg\min_{\boldsymbol{\beta}_j} \|\left[\mathbf{Y}_{(3)}^T\right]_{:,j} - \left[\mathbf{Y}_{(3)}^T\right]_{\partial_j} \boldsymbol{\beta}_j\|_F^2 \\ &= \left(\left[\mathbf{Y}_{(3)}^T\right]_{\partial_j}^T \left[\mathbf{Y}_{(3)}^T\right]_{\partial_j}\right)^{-1} \left[\mathbf{Y}_{(3)}^T\right]_{\partial_j}^T \left[\mathbf{Y}_{(3)}^T\right]_{:,j}.\end{aligned} \tag{7}$$

Given $\hat{\boldsymbol{\beta}}_j$, the estimate of regression error, $\hat{\boldsymbol{\xi}}_j$, is computed by

$$\hat{\boldsymbol{\xi}}_j = \left[\mathbf{Y}_{(3)}^T\right]_{:,j} - \left[\mathbf{Y}_{(3)}^T\right]_{\partial_j} \hat{\boldsymbol{\beta}}_j. \tag{8}$$

Next, we introduce two strategies to estimate uniform eigennoise level map for spatially invariant noise and nonuniform eigennoise level maps for spatially variant noise, respectively.

1) Uniform Eigennoise Level Map: This section considers the case that noise is spatially invariant. The standard deviation of Gaussian noise in jth band, denoted by $\hat{\sigma}_j$, is estimated by computing

$$\hat{\sigma}_j = \text{std}\left(\hat{\boldsymbol{\xi}}_j\right) \tag{9}$$

where $\text{std}(\cdot)$ is a function computing standard deviation of the elements of the input vector.

Finally, we obtain an estimate of noise spectral covariance, denoted by $\widehat{\mathbf{C}} = diag(\hat{\sigma}_1^2, \hat{\sigma}_2^2, \cdots, \hat{\sigma}_b^2) \in \mathbb{R}^{b\times b}$, where $\hat{\sigma}_j^2$ represents the estimated noise variance of jth band. The noise spectral covariance is diagonal, implying that we consider a simple case where noise is both pixelwise and bandwise independent. Consequently, we can estimate noise level of observed eigenimages (denoted by $\widehat{\widetilde{\mathbf{C}}} \in \mathbb{R}^{p\times p}$) as follows:

$$\widehat{\widetilde{\mathbf{C}}} = \mathbf{E}^T \widehat{\mathbf{C}} \mathbf{E} = \text{diag}\left(\hat{\tilde{\sigma}}_1^2, \hat{\tilde{\sigma}}_2^2, \cdots, \hat{\tilde{\sigma}}_p^2\right) \tag{10}$$

where $\hat{\tilde{\sigma}}_i^2$ represents the noise variance of ith eigenimage.

The utilization of noise level can significantly enhance the flexibility of our pretrained denoising neural network, enabling it to effectively adapt to new images with varying noise intensities. Since the dimension of noise level $\tilde{\sigma}_i$ is inconsistent with that of the eigenimage $\widetilde{\mathbf{Z}}_i$, we generate a eigennoise level map (denoted as $\widetilde{\mathbf{O}}_i \in \mathbb{R}^{r\times c}$), i.e.,

$$\widetilde{\mathbf{O}}_i = \hat{\tilde{\sigma}}_i \mathbf{1}_r \mathbf{1}_c^T \tag{11}$$

where $\mathbf{1}_r \in \mathbb{R}^r$ denote an all-ones column vector whose dimension is given by the subscript.

2) Nonuniform Eigennoise Level Map: This section introduces the strategy to estimate nonuniform eigennoise level maps when noise is spatially variant. Given the estimate of regression error $\hat{\boldsymbol{\xi}}_j$, we generate a tensor of nonuniform eigennoise level maps as follows:

$$\widetilde{\mathcal{O}} = (\Xi \odot \Xi) \times_3 (\mathbf{E} \odot \mathbf{E}) \tag{12}$$

where $\odot$ denotes elementwise multiplication, $\Xi \in \mathbb{R}^{r\times c\times b}$ is the tensor version of regression errors, e.g., $\text{vec}(\Xi(:,:,j)) = \hat{\boldsymbol{\xi}}_j$, and $(\Xi \odot \Xi)$ represents a coarse estimate of the variance of spatially variant noise. The derivation of (12) is inspired by the characteristic of Gaussian distributed data, where the sum of multiple Gaussian distributions still satisfies a Gaussian probability distribution (for detailed derivation, refer to [61, Supplement S1 Supplement S1]).

IV. Proposed HSI Denoising Network: Eigen-CNN

We introduce an HSI denoising framework Eigen-CNN, which represents the HSI in a low-dimensional subspace and mitigates noise in the eigenimages (i.e., representation coefficients) using a eigenimage-wise restoration network. Then

[1]Code available at: https://github.com/LinaZhuang/HSI-MixedNoiseRemoval-FastHyMix.

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network learning is guided by the first eigenimage and noise images.

Given $\mathbf{E}$, the HSI denoising problem is formulated as follows:

$$\widehat{\mathcal{Z}} \in \arg\min_{\mathcal{Z}} \frac{1}{2}\|\mathcal{Y} - \mathcal{Z} \times_3 \mathbf{E}\|_F^2 + \lambda\phi_{CNN}(\mathcal{Z}) = \arg\min_{\mathbf{Z}} \frac{1}{2}\|\mathcal{Y} \times_3 \mathbf{E}^T - \mathcal{Z}\|_F^2 + \lambda\phi_{CNN}(\mathcal{Z}). \quad (13)$$

The first term on the right-hand of side (13) represents the data fidelity and accounts for the zero-mean Gaussian noise, while the second term is a regularizer, ϕ_{CNN}, expressing prior information of eigenimages with a CNN network. We remark that (13) is the so-called denoising operator, or Moreau proximity operator of ϕ_{CNN} [66]. Resorting to plug-and-play trick [67], we introduce an eigenimage-guided CNN network to estimate eigenimages, i.e.,

$$\widehat{\mathcal{Z}} \leftarrow f_\theta\left(\widetilde{\mathcal{Z}}, \lambda\right) \quad (14)$$

where $\widetilde{\mathcal{Z}} = \mathcal{Y} \times_3 \mathbf{E}^T$, $f_\theta(\cdot)$ is the eigenimage-guided CNN network parameterized by θ, and parameter λ is related to standard deviation of Gaussian noise in the input eigenimages.

Next, we introduce our eigenimage-guided CNN network, which consists of inference part and training part.

A. Inference Part

The inference part of our Eigen-CNN framework is depicted in Fig. 1. The proposed restoration process consists of three steps.

Step 1: Generate observed eigenimages by projecting the noisy HSI $\mathcal{Y}$ onto a prelearned orthogonal subspace: $\widetilde{\mathcal{Z}} = \mathcal{Y} \times_3 \mathbf{E}^T$.

Step 2: Remove noise in the observed eigenimages using an eigenimage-wise network

$$\widehat{\mathbf{Z}}_i = f_\theta\left(\text{cat}\left(\widetilde{\mathbf{Z}}_1, \widetilde{\mathbf{Z}}_i, \widetilde{\mathbf{O}}_i\right)\right), \quad (i = 1, 2, \cdots, p) \quad (15)$$

where $\widetilde{\mathbf{Z}}_i := \widetilde{\mathcal{Z}}(:, :, i)$ and $\widetilde{\mathbf{O}}_i := \widetilde{\mathcal{O}}(:, :, i)$ represent ith noisy eigenimage and ith noise map, respectively. Function cat$(\mathbf{A}, \mathbf{B})$ concatenates matrix $\mathbf{B}$ to the end of matrix $\mathbf{A}$ along dimension 3. We can see from (15) that the network aims to estimate the clean ith eigenimage (denoted by $\widehat{\mathbf{Z}}_i := \widehat{\mathcal{Z}}(:, :, i)$) while $(\widetilde{\mathbf{Z}}_1, \widetilde{\mathbf{O}}_i)$ serve as guide images in the denoising network.

Step 3: An underlying clean HSI can be estimated by computing

$$\widehat{\mathcal{X}} = \widehat{\mathcal{Z}} \times_3 \mathbf{E}. \quad (16)$$

B. Training Part

The training part of Eigen-CNN is introduced in this section. The eigenimage network was trained using a large amount of noisy-clean eigenimage pairs

$$\min_\theta \sum_i \mathbb{E}_{(\widetilde{\mathbf{Z}}_1, \widetilde{\mathbf{Z}}_i, \mathbf{Z}_i)} \| f_\theta\left(\text{cat}\left(\widetilde{\mathbf{Z}}_1, \widetilde{\mathbf{Z}}_i, \widetilde{\mathbf{O}}_i\right)\right) - \mathbf{Z}_i \|_F^2 \quad (17)$$

where cat$(\widetilde{\mathbf{Z}}_1, \widetilde{\mathbf{Z}}_i, \widetilde{\mathbf{O}}_i)$ is input data, and clean eigenimage $\mathbf{Z}_i$ is the target data for training.

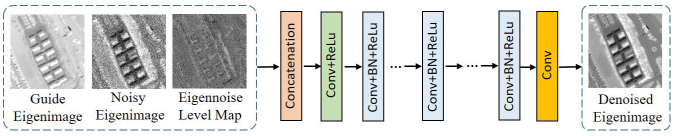


Fig. 2. Architecture of the proposed eigenimages plus eigennoise level maps guided network.

TABLE I
OUR NETWORK CONFIGURATION FOR EIGENIMAGE DENOISING

Concatenation of guided eigenimage and eigennoise level map					
Layer	Type	Input data			Output data size
1 (input)	Concatenation	$\{\widetilde{\mathbf{Z}}_1, \widetilde{\mathbf{Z}}_i, \widetilde{\mathbf{O}}_i\} \in \mathbb{R}^{r\times c}$			$r \times c \times 3$
Convolutional mapping					
Layer	Type	Kernel size	Number of Kernels	Input data size	Output data size
2	Conv+ReLU	3×3	64	$r \times c \times 3$	$r \times c \times 64$
3 - 17	Conv+BN+ReLU	3×3	64	$r \times c \times 64$	$r \times c \times 64$
18 (output)	Conv	3×3	1	$r \times c \times 64$	$r \times c \times 1$

A deep CNN architecture is adopted to learn the mapping relationship from the input noisy eigenimages to the output denoised eigenimage directly through a multilayer convolutional network structure [68]. Table I lists the details of our network configuration, which is depicted in Fig. 2. The network comprises 18 layers. Layer 1 concatenate the input data (namely, the first eigenimage $\widetilde{\mathbf{Z}}_1$, the ith eigenimage $\widetilde{\mathbf{Z}}_i$, and the ith eigennoise level map $\widetilde{\mathbf{O}}_i$), resulting an output data of size $r \times c \times 3$. Layer 2 is a convolutional layer with 64 filters of size 3×3, a stride of 1, and padding of 1, followed by a nonlinear activation function rectified linear unit (ReLU). The ReLU activation function enhances the network's nonlinear expressive power. Layers 3–17 are the same, consisting of a convolutional layer, a batch normalization layer, and a nonlinear activation function (ReLU). Each convolutional layer contains 64 filters of size 3×3, a stride of 1, and padding of 1. The batch normalization layer is used to accelerate training and improve model generalization capabilities. Layer 18 is a convolutional layer with one filter size of 3×3, a stride of 1, and padding of 1. The output layer generates the denoised eigenimage. This structure maintains network depth while reducing the number of model parameters, thereby lowering the risk of overfitting. However, we emphasize that we have merely employed common convolutional modules and that the network's architecture is straightforward and replaceable by more complex and advanced structures.

V. EXPERIMENTS ON SIMULATED HSIS

A. Implementation Details

1) Simulated Data on ICVL Data for Network Training: Our network was trained using the Interdisciplinary Computational Vision Laboratory (ICVL) hyperspectral dataset[2] [69], which was acquired by a ground-based camera (Specim PS Kappa DX in the ICVL). The ICVL dataset consists 201 images collected at 1392×1300 spatial pixels with 31 spectral bands. The ICVL images have high SNR. Thus,

[2]http://icvl.cs.bgu.ac.il/hyperspectral/

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(a) (b) (c) (d)

Fig. 3. HSIs used in the experiments. (a) Subimage of Washington DC Mall data. (b) Subimage of Pavia University data. (c) Hyperion River. (d) Hyperion Cuprite.

we simulated noisy HSIs by adding independent Gaussian noise as follows. Let $\mathbf{o}_i \in \mathbb{R}^b$ be a Gaussian noise vector of the ith spectral pixel. We simulated pixelwise and bandwise independent Gaussian noise: $\mathbf{o}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{D}^2)$, where $\mathcal{N}$ is a zero-mean Gaussian distribution, and $\mathbf{D}$ is a diagonal matrix with diagonal elements $d_{ii} = \alpha_1 \text{median}(\mathcal{X}(:,:,i))$. We set $\alpha_1 = 0.2$. We set the Gaussian noise intensity of each band according to the median value per band so that corresponding eigenimages with diverse Gaussian noise intensities can be used to train a flexible denoising network. Noisy eigenimages can be obtained from the noisy ICVL images using (3) and (4). Eigenimages obtained by randomly selected 90% of ICVL images, are used for network training. The remaining 10% ICVL images are used for network validation. The subspace dimension is set to 3, and it is sufficient to represent spectral vectors well.

Our network, pretrained on ICVL eigenimages, can be directly applied to work on real HSIs without fine-tuning because the pretrained network has been quite flexible to diverse noise intensities. Our pretrained network has been directly applied to test HSI images, whose noise simulation is introduced in Section V-A2.

2) Simulated Noisy Testing HSI Datasets: We use two datasets of HSIs: the subreigon of Washington DC Mall data (of size $[130 \times 130]$ (pixels) $\times$ 100 (bands)) and the subregion of Pavia University data (of size $[250 \times 250]$ (pixels) $\times$ 87 (bands)), which are high-quality [as shown in Fig. 3 (a) and (b)] and considered as clean images in experiments. Gaussian noise, $\mathbf{o}_i \sim \mathcal{N}(\mathbf{0}, \mathbf{D}^2)$, with $\alpha \in \{0.42, 0.20, 0.10, 0.05\}$ was added to simulate noisy Washington dc Mall data in cases 1–4. Gaussian noise with $\alpha \in \{0.90, 0.50, 0.35, 0.15\}$ was added to simulate noisy Pavia University data in cases 5–8. Note that we did not simulate stripe noise or impulse noise because of the Gaussian noise assumption in our model. For all simulated noisy HSIs, we empirically set the subspace dimension to six. The impact of subspace dimension on the performance of our denoising network is analyzed in Section V-D.

3) Competing Methods: We compare our method with state-of-the-art model-based/machine learning-based methods, namely, NonRLRS [30], KBR [70], LRTF-DFR[3] [19], NG-meet[4] [41], and FastHyMix[5] [31]. All experiments were implemented in MATLAB (R2022a) on Windows 10 with an ADM Ryzen 9 4900HS 3.00-GHz processor and 16-GB RAM. NonRLRS [30], KBR [70] are sparse and low-rank representation methods. It is not necessary to set subspace dimension numbers. Here, their parameters are fine tuned to get optimal results for simulated HSIs, see Tables II, III, and IV. LRTF-DFR [19], NG-meet [41], and FastHyMix [31] are machine-learning-based methods. It is required to set the subspace dimension number. Here, we report the performance of these methods using different numbers ($p \in \{6, 8, 10, 12\}$) of subspace dimension, see Tables II–VIII. In additional, deep learning-based methods (including QRNN3D[6] [60], HSID-CNN[7] [58], NL-3DCNN [71], and MAC-Net[8] [72]) are compared. Note that networks of QRNN3D and HSID-CNN are working on the original image domain, whereas the NL-3DCNN and MAC-Net networks are denoising eigenimages (meaning only NL-3DCNN and MAC-Net are required to set subspace dimension number). We present the denoising results by QRNN3D and HSID-CNN in Tables II, III, and IV. However, NL-3DCNN and MAC-Net are flexible in the testing/inference procedure for different values of subspace dimension. Here, the subspace dimension number is validated and set to be six in their training for the ICVL images. In the testing/inference procedure, we test simulated HSIs using different numbers ($p \in \{6, 8, 10, 12\}$) of subspace dimension, see Tables II–VIII.

NonRLRS and KBR are both state-of-the-art low-rank and sparse representation-based methods for HSI denoising. The former decomposes a noisy HSI, expressed in a matrix form, into low-rank clean HSI and sparse noise components. The latter utilizes sparse representation (i.e., Kronecker-basis-representation) of the underlying HSI. LRTF-DFR, NG-meet, FastHyMix, NL-3DCNN, MAC-Net, and Eigen-CNN belong to the same category of methods. These methods all involve eigendecomposition to perform denoising on eigenimages rather than in the original image space. The difference lies in how they remove noise from eigenimages. For example, the LRTF-DFR method minimizes the total variation in the spatial dimension of eigenimages. FastHyMix characterizes the complex distribution of mixed noise by using a Gaussian mixture model, resulting in eigenimages containing very little Gaussian noise. NL-3D CNN uses a 3-D CNN network to denoise similar cubes in eigenimages. MAC-Net denoises eigenimages using a 3-D residual U-Net. Eigen-CNN uses eigennoise level maps to guide the network in further removing residual noise from eigenimages. The aforementioned methods work on the orthogonally transformed domain. However, HSID-CNN and QRNN3D networks work on the original image domain. HSID-CNN takes as input a single spectral band along with its neighboring bands, with the goal of removing noise from the central band. QRNN3D utilizes 3-D convolution to simultaneously extract both spatial and spectral features.

4) Evaluation Metrics: We use three well-known metrics measuring image quality: the peak SNR (PSNR) index, the structural similarity (SSIM) index, and the feature similarity

[3]https://yubangzheng.github.io/homepage/#publications
[4]https://github.com/quanmingyao/NGmeet
[5]https://github.com/LinaZhuang/HSI-MixedNoiseRemoval-FastHyMix
[6]https://github.com/Vandermode/QRNN3D
[7]https://github.com/qzhang95/HSID-CNN
[8]https://github.com/bearshng/mac-net

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TABLE II

QUANTITATIVE ASSESSMENT OF DIFFERENT DENOISING METHODS APPLIED TO SIMULATED WASHINGTON DC MALL IMAGES CORRUPTED BY VARYING INTENSITY OF GAUSSIAN NOISE

		Noisy	NonRLRS (TIP, 2020) [30]	KBR (TPAMI, 2018) [70]	LRTF-DFR (TGRS, 2020) ($p = 6$) [19]	NG-meet (TPAMI, 2020) ($p = 6$) [41]	FastHyMix (TNNLS, 2021) ($p = 6$) [31]	HSID-CNN (TRGS, 2018) [58]	QRNN3D (TNNLS, 2020) [60]	NL-3DCNN (TGRS, 2022) ($p = 6$) [71]	MAC-Net (TGRS, 2022) ($p = 6$) [72]	Eigen-CNN (proposed) ($p = 6$)
Case 1	MPSNR (dB)	16.40	23.65	28.86	28.75	31.90	33.49	28.18	26.89	34.08	33.07	**34.40**
	MSSIM	0.47005	0.8796	0.9157	0.8924	0.9486	0.9667	0.9125	0.8828	0.9711	0.9619	**0.9752**
	MFSIM	0.72433	0.9212	0.9458	0.9346	0.9648	0.9732	0.9421	0.9256	0.9770	0.9709	**0.9793**
Case 2	MPSNR (dB)	22.84	29.48	34.40	35.88	37.67	38.31	31.27	30.33	38.85	38.20	**39.04**
	MSSIM	0.7379	0.9544	0.9693	0.9756	0.9854	0.9871	0.957	0.9494	0.9888	0.9865	**0.9901**
	MFSIM	0.8599	0.9670	0.9784	0.9819	0.9883	0.9891	0.969	0.9658	0.9904	0.9888	**0.9913**
Case 3	MPSNR (dB)	28.86	35.41	39.40	41.95	42.94	43.14	32.80	31.37	43.52	40.46	**43.59**
	MSSIM	0.8976	0.9860	0.9899	0.9948	0.9954	0.9953	0.9705	0.9603	0.9958	0.9930	**0.9961**
	MFSIM	0.9392	0.9881	0.9921	0.9951	0.9959	0.9957	0.978	0.9742	0.9960	0.9942	**0.9963**
Case 4	MPSNR (dB)	34.88	41.01	44.43	45.42	47.92	47.77	33.36	31.65	**47.89**	40.44	47.82
	MSSIM	0.9685	0.9950	0.9972	0.9982	0.9985	0.9984	0.9743	0.9627	0.9985	0.9762	**0.9986**
	MFSIM	0.9790	0.9949	0.9972	0.9978	0.9984	0.9981	0.9806	0.9762	0.9981	0.9964	**0.9982**

Note: Best results are highlighted in bold.
p denotes the setting of subspace dimension used in LRTF-DFR, NG-meet, FastHyMix, NL-3DCNN, MAC-Net, and Eigen-CNN.

TABLE III

QUANTITATIVE ASSESSMENT OF DIFFERENT DENOISING METHODS APPLIED TO SIMULATED PAVIA UNIVERSITY IMAGES CORRUPTED BY VARYING INTENSITY OF GAUSSIAN NOISE

		Noisy	NonRLRS (TIP, 2020) [30]	KBR (TPAMI, 2018) [70]	LRTF-DFR (TGRS, 2020) ($p = 6$) [19]	NG-meet (TPAMI, 2020) ($p = 6$) [41]	FastHyMix (TNNLS, 2021) ($p = 6$) [31]	HSID-CNN (TRGS, 2018) [58]	QRNN3D (TNNLS, 2020) [60]	NL-3DCNN (TGRS, 2022) ($p = 6$) [71]	MAC-Net (TGRS, 2022) ($p = 6$) [72]	Eigen-CNN (proposed) ($p = 6$)
						$p = 6$						
Case 5	MPSNR (dB)	17.87	26.70	31.25	29.98	34.41	36.85	32.71	30.64	37.04	35.85	**37.51**
	MSSIM	0.2557	0.8055	0.8518	0.7808	0.9064	0.9596	0.9089	0.8193	0.9579	0.9396	**0.9641**
	MFSIM	0.5900	0.8987	0.9349	0.9000	0.9596	0.9734	0.9414	0.9133	0.9752	0.9677	**0.9764**
Case 6	MPSNR (dB)	22.98	31.06	35.96	35.22	38.98	40.15	35.85	34.71	40.47	39.59	**40.60**
	MSSIM	0.4732	0.9131	0.9354	0.9001	0.9639	0.9779	0.9562	0.9312	0.9785	0.9727	**0.9804**
	MFSIM	0.7534	0.9539	0.9721	0.9609	0.9837	0.9866	0.9705	0.9647	0.9880	0.9855	**0.9881**
Case 7	MPSNR (dB)	26.08	34.28	37.88	37.96	41.54	42.22	37.41	36.47	42.52	41.76	**42.54**
	MSSIM	0.6160	0.9514	0.9619	0.9464	0.9792	0.9851	0.9705	0.9595	0.9858	0.9828	**0.9866**
	MFSIM	0.8356	0.9741	0.9833	0.9791	0.9907	0.9915	0.9805	0.978	**0.9924**	0.9911	**0.9924**
Case 8	MPSNR (dB)	33.44	41.15	40.81	45.09	47.48	47.21	39.77	38.61	**47.46**	37.76	47.11
	MSSIM	0.8704	0.9870	0.9869	0.9933	0.9944	0.9945	0.9846	0.9814	**0.9949**	0.9833	0.9947
	MFSIM	0.9524	0.9929	0.9939	0.9967	0.9976	0.9972	0.9907	0.9888	**0.9975**	0.9932	0.9974

Note: Best results are highlighted in bold.
p denotes the setting of subspace dimension used in LRTF-DFR, NG-meet, FastHyMix, NL-3DCNN, MAC-Net, and Eigen-CNN.

TABLE IV

QUANTITATIVE ASSESSMENT OF DIFFERENT DENOISING METHODS APPLIED TO SIMULATED WASHINGTON DC MALL DATA AND PAVIA UNIVERSITY DATA CORRUPTED BY SPATIALLY VARIANT GAUSSIAN NOISE

		Noisy	NonRLRS (TIP, 2020) [30]	KBR (TPAMI, 2018) [70]	LRTF-DFR (TGRS, 2020) ($p = 6$) [19]	NG-meet (TPAMI, 2020) ($p = 6$) [41]	FastHyMix (TNNLS, 2021) ($p = 6$) [31]	HSID-CNN (TGRS 2018) [58]	QRNN3D (TNNLS, 2020) [60]	NL-3DCNN (TGRS, 2022) ($p = 6$) [71]	MAC-Net (TGRS, 2022) ($p = 6$) [72]	Eigen-CNN (proposed) ($p = 6$)
						Washington DC Mall data						
Case 9	MPSNR (dB)	19.25	26.81	32.07	33.13	34.26	35.62	30.00	29.10	35.94	35.43	**36.80**
	MSSIM	0.6120	0.9324	0.9512	0.9570	0.9692	0.9790	0.9425	0.9293	0.9803	0.9781	**0.9861**
	MFSIM	0.7900	0.9504	0.9670	0.9700	0.9779	0.9829	0.9595	0.9522	0.9840	0.9824	**0.9874**
						Pavia University data						
Case 10	MPSNR (dB)	18.24	27.12	32.14	30.93	34.77	36.19	33.04	31.44	36.44	35.85	**37.50**
	MSSIM	0.2926	0.8454	0.8871	0.8265	0.9269	0.9586	0.9251	0.8629	0.9597	0.9496	**0.9695**
	MFSIM	0.6129	0.9108	0.9479	0.9268	0.9655	0.9736	0.9504	0.9283	0.9752	0.9714	**0.9801**

Note: Best results are highlighted in bold.
p denotes the setting of subspace dimension used in LRTF-DFR, NG-meet, FastHyMix, NL-3DCNN, MAC-Net, and Eigen-CNN.

(FSIM) index. We calculate the indexes for each band, and then take the average of all the indexes across the bands, reporting them in Tables II–IX.

The mean PSNR (MPSNR) is defined as follows:

$$\text{MPSNR} = \frac{1}{b}\sum_{i=1}^{b} 10\log_{10}\frac{\max(\mathcal{X}(:,:,i))^2}{\|\mathcal{X}(:,:,i) - \widehat{\mathcal{X}}(:,:,i)\|_F^2} \tag{18}$$

where $\max(\mathbf{A})$ returns the largest element in matrix $\mathbf{A}$. Symbols $\mathcal{X}(:,:,i)$ and $\widehat{\mathcal{X}}(:,:,i)$ represent the clean ith band and the estimated ith band, respectively.

The mean SSIM (MSSIM) is calculated as follows:

$$\text{MSSIM} = \frac{1}{b}\sum_{i=1}^{b} \text{SSIM}\Big(\mathcal{X}(:,:,i), \widehat{\mathcal{X}}(:,:,i)\Big) \tag{19}$$

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TABLE V

QUANTITATIVE ASSESSMENT OF EIGENIMAGE-BASED METHODS (WITH DIFFERENT SETTING OF SUBSPACE DIMENSION) APPLIED TO SIMULATED WASHINGTON DC MALL IMAGES CORRUPTED BY VARYING INTENSITY OF GAUSSIAN NOISE

		Noisy	LRTF-DFR (TGRS, 2020) [19]	NG-meet (TPAMI, 2020) [41]	FastHyMix (TNNLS, 2021) [31]	NL-3DCNN (TGRS, 2022) [71]	MAC-Net (TGRS, 2022) [72]	Eigen-CNN (proposed)
					$p = 8$			
Case 1	MPSNR (dB)	16.40	27.91	29.18	33.46	34.11	32.77	**34.38**
	MSSIM	0.47005	0.8667	0.9046	0.9666	0.9713	0.9594	**0.9751**
	MFSIM	0.72433	0.9218	0.9426	0.9731	0.9772	0.9687	**0.9792**
Case 2	MPSNR (dB)	22.84	34.91	35.32	38.30	38.93	38.04	**39.04**
	MSSIM	0.7379	0.9673	0.9724	0.9871	0.9890	0.9858	**0.9901**
	MFSIM	0.8599	0.9763	0.9807	0.9891	0.9905	0.9883	**0.9913**
Case 3	MPSNR (dB)	28.86	41.62	41.06	43.27	**43.77**	40.31	43.75
	MSSIM	0.8976	0.9935	0.9919	0.9953	0.9959	0.9930	**0.9962**
	MFSIM	0.9392	0.9944	0.9938	0.9959	0.9964	0.9944	**0.9966**
Case 4	MPSNR (dB)	34.88	45.86	46.70	48.37	**48.65**	37.18	48.48
	MSSIM	0.9685	0.9982	0.9977	0.9985	**0.9986**	0.9949	**0.9986**
	MFSIM	0.9790	0.9979	0.9981	0.9985	0.9986	0.9956	**0.9987**
					$p = 10$			
Case 1	MPSNR (dB)	16.40	27.23	27.19	33.44	34.11	32.45	**34.36**
	MSSIM	0.47005	0.8446	0.8625	0.9665	0.9712	0.9564	**0.9751**
	MFSIM	0.72433	0.9107	0.9212	0.9730	0.9772	0.9667	**0.9792**
Case 2	MPSNR (dB)	22.84	34.40	33.54	38.29	38.94	37.85	**39.03**
	MSSIM	0.7379	0.9593	0.9582	0.9871	0.9890	0.9852	**0.9901**
	MFSIM	0.8599	0.9717	0.9726	0.9891	0.9906	0.9878	**0.9913**
Case 3	MPSNR (dB)	28.86	41.30	39.39	43.25	**43.81**	39.62	43.74
	MSSIM	0.8976	0.9929	0.9879	0.9953	0.9960	0.9927	**0.9962**
	MFSIM	0.9392	0.9938	0.9912	0.9959	0.9964	0.9943	**0.9966**
Case 4	MPSNR (dB)	34.88	45.86	45.19	48.37	**48.73**	37.18	48.48
	MSSIM	0.9685	0.9982	0.9967	0.9985	**0.9986**	0.9948	**0.9986**
	MFSIM	0.9790	0.9980	0.9974	0.9986	**0.9987**	0.9956	**0.9987**
					$p = 12$			
Case 1	MPSNR (dB)	16.40	26.82	25.95	33.42	34.11	32.14	**34.34**
	MSSIM	0.47005	0.8316	0.8312	0.9664	0.9712	0.9533	**0.9750**
	MFSIM	0.72433	0.9038	0.9058	0.9730	0.9772	0.9641	**0.9792**
Case 2	MPSNR (dB)	22.84	33.95	32.33	38.27	38.94	37.59	**39.02**
	MSSIM	0.7379	0.9546	0.9465	0.9870	0.9890	0.9843	**0.9901**
	MFSIM	0.8599	0.9683	0.9657	0.9890	0.9905	0.9870	**0.9912**
Case 3	MPSNR (dB)	28.86	41.13	38.26	43.24	**43.80**	39.16	43.73
	MSSIM	0.8976	0.9925	0.9846	0.9953	0.9960	0.9924	**0.9962**
	MFSIM	0.9392	0.9934	0.9890	0.9959	0.9964	0.9941	**0.9966**
Case 4	MPSNR (dB)	34.88	45.93	44.15	48.36	**48.73**	36.78	48.48
	MSSIM	0.9685	0.9983	0.9959	0.9985	**0.9986**	0.9945	**0.9986**
	MFSIM	0.9790	0.9980	0.9968	0.9986	**0.9987**	0.9954	**0.9987**

Note: Best results are highlighted in bold.
p denotes the setting of subspace dimension used in LRTF-DFR, NG-meet, FastHyMix, NL-3DCNN, MAC-Net, and Eigen-CNN.

where SSIM($\mathbf{A}, \mathbf{B}$) calculate the SSIM index of image $\mathbf{A}$ and image $\mathbf{B}$. The SSIM metric, defined in [73], measures the similarity between two images and is considered to be correlated with the quality perception of the human visual system (HVS).

The mean FSIM (MFSIM) is obtained as follows:

$$\text{MFSIM} = \frac{1}{b}\sum_{i=1}^{b} \text{FSIM}\left(\mathcal{X}(:,:,i), \widehat{\mathcal{X}}(:,:,i)\right) \tag{20}$$

where FSIM($\mathbf{A}$ and $\mathbf{B}$) calculate the FSIM index of image $\mathbf{A}$ and image $\mathbf{B}$. The FSIM metric, defined in [74], measures the similarity of images using gradient magnitude and Fourier phase congruency.

B. Experimental Results on Simulated Testing Datasets

Two simulated noisy datasets, namely, Washington DC Mall dataset and Pavia University dataset, was tested. We remark that experiments in this article were conducted on small subimages, and the experimental results from testing on the complete large images have been made available at the external link[9] for readers' reference. Denoising performance of each method on two datasets in terms of MPSNR, MSSIM, and MFSIM is presented in Tables II and III, where we can observe that the proposed Eigen-CNN and NL-3DCNN method achieved the first and second highest denoising accuracy, respectively. Note that in traditional machine-learning methods (LRTF-DFR, NG-meet, and FastHyMix) and deep learning methods (Eigen-

[9]https://www.dropbox.com/scl/fi/frr4v72ltonp6mkf8pmap/Eigen_CNN_exp_large_imgs.pdf?rlkey=lc9xm 4hesjpvipind9qxjzbnm&dl=0

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TABLE VI

QUANTITATIVE ASSESSMENT OF EIGENIMAGE-BASED METHODS (WITH DIFFERENT SETTING OF SUBSPACE DIMENSION) APPLIED TO SIMULATED PAVIA UNIVERSITY IMAGES CORRUPTED BY VARYING INTENSITY OF GAUSSIAN NOISE

		Noisy	LRTF-DFR (TGRS, 2020) [19]	NG-meet (TPAMI, 2020) [41]	FastHyMix (TNNLS, 2021) [31]	NL-3DCNN (TGRS, 2022) [71]	MAC-Net (TGRS, 2022) [72]	Eigen-CNN (proposed)
					$p = 8$			
Case 5	MPSNR (dB)	17.87	29.51	30.84	36.83	37.05	35.31	**37.48**
	MSSIM	0.2557	0.7601	0.8033	0.9594	0.9580	0.9288	**0.9639**
	MFSIM	0.5900	0.8864	0.9220	0.9733	0.9752	0.9639	**0.9763**
Case 6	MPSNR (dB)	22.98	34.63	35.78	40.13	40.46	39.26	**40.58**
	MSSIM	0.4732	0.8750	0.9189	0.9778	0.9786	0.9690	**0.9803**
	MFSIM	0.7534	0.9499	0.9690	0.9865	0.9880	0.9843	**0.9881**
Case 7	MPSNR (dB)	26.08	37.28	38.73	42.24	42.55	41.55	**42.56**
	MSSIM	0.6160	0.9264	0.9535	0.9851	0.9859	0.9811	**0.9866**
	MFSIM	0.8356	0.9713	0.9824	0.9915	**0.9924**	0.9906	**0.9924**
Case 8	MPSNR (dB)	33.44	45.40	45.58	47.50	**47.83**	37.22	47.46
	MSSIM	0.8704	0.9924	0.9889	0.9947	**0.9951**	0.9807	0.9949
	MFSIM	0.9524	0.9964	0.9958	0.9974	**0.9977**	0.9920	0.9975
					$p = 10$			
Case 5	MPSNR (dB)	17.87	29.18	28.61	36.79	37.05	34.83	**37.46**
	MSSIM	0.2557	0.7462	0.7212	0.9591	0.9580	0.9202	**0.9638**
	MFSIM	0.5900	0.8760	0.8872	0.9732	0.9752	0.9602	**0.9763**
Case 6	MPSNR (dB)	22.98	34.24	33.64	40.11	40.46	38.90	**40.57**
	MSSIM	0.4732	0.8580	0.8755	0.9778	0.9786	0.9659	**0.9803**
	MFSIM	0.7534	0.9416	0.9534	0.9865	0.9880	0.9831	**0.9881**
Case 7	MPSNR (dB)	26.08	36.83	36.67	42.22	**42.56**	41.28	42.55
	MSSIM	0.6160	0.9113	0.9284	0.9851	0.9859	0.9796	**0.9866**
	MFSIM	0.8356	0.9650	0.9737	0.9915	**0.9924**	0.9900	**0.9924**
Case 8	MPSNR (dB)	33.44	45.79	43.78	47.50	**47.83**	37.01	47.45
	MSSIM	0.8704	0.9920	0.9833	0.9947	**0.9951**	0.9795	0.9949
	MFSIM	0.9524	0.9962	0.9940	0.9974	**0.9977**	0.9915	0.9975
					$p = 12$			
Case 5	MPSNR (dB)	17.87	29.02	27.26	36.76	37.04	34.35	**37.44**
	MSSIM	0.2557	0.7404	0.6655	0.9589	0.9580	0.9105	**0.9637**
	MFSIM	0.5900	0.8708	0.8607	0.9732	0.9752	0.9557	**0.9762**
Case 6	MPSNR (dB)	22.98	33.97	32.32	40.09	40.46	38.53	**40.56**
	MSSIM	0.4732	0.8464	0.8418	0.9777	0.9786	0.9625	**0.9803**
	MFSIM	0.7534	0.9354	0.9401	0.9865	0.9880	0.9816	**0.9881**
Case 7	MPSNR (dB)	26.08	36.53	35.36	42.20	**42.56**	40.99	42.54
	MSSIM	0.6160	0.8999	0.9075	0.9850	0.9859	0.9779	**0.9866**
	MFSIM	0.8356	0.9599	0.9662	0.9915	**0.9924**	0.9893	**0.9924**
Case 8	MPSNR (dB)	33.44	45.76	42.58	47.49	**47.83**	36.63	47.44
	MSSIM	0.8704	0.9915	0.9789	0.9947	**0.9951**	0.9775	0.9949
	MFSIM	0.9524	0.9960	0.9925	0.9974	**0.9977**	0.9907	0.9975

Note: Best results are highlighted in bold.
p denotes the setting of subspace dimension used in LRTF-DFR, NG-meet, FastHyMix, MAC-Net, NL-3DCNN, and Eigen-CNN.

CNN, NL-3DCNN, and MAC-Net), the subspace dimensional number is set to be 6 for comparison. Figs. 4 and 5 present the denoised results of one band for each method. Fig. 6 depicts the PSNR results on each band. In dealing with Gaussian noise of different intensities, the proposed Eigen-CNN method achieves the best results in most cases (apart from in low-noise cases). The superiority of our method becomes more pronounced as the intensity of the noise increases. Moreover, the training and testing computation time required by Eigen-CNN are significantly less than those of NL-3DCNN. Note that NL-3DCNN requires to train 3-D cubes and search for similarity cubes in the network, and Eigen-CNN only requires to train 2-D image patches. For reference, the average training time by Eigen-CNN is 8 h in MATLAB while the average training time by NL-3DCNN is 15 h in PyTorch; the average testing time by Eigen-CNN is 2 s while the average testing time by NL-3DCNN is 130 s.

Next, we discuss the results of deep learning-based competitors (namely, HSID-CNN, QRNN3D, NL-3DCNN, MAC-Net, and Eigen-CNN). Note that all deep learning-based competitors were all pre-trained using the same data, i.e., 180 paired noisy-clean ICVL images. We can see that Eigen-CNN, NL-3DCNN, and MAC-Net outperform the other two significantly. Despite the differences in network structures among these deep learning-based methods. We guess that the improved accuracy of Eigen-CNN, NL-3DCNN, and MAC-Net can be primarily attributed to its eigenimage-based denoising strategy. HSID-CNN and QRNN3D networks work on the original image domain, whereas Eigen-CNN, NL-3DCNN, and MAC-Net work on the orthogonally transformed domain, leading to a better noise suppression. This strategy can effectively exploit the high spectral correlation in hyperspectral data, and has proven to be effective in traditional machine learning-based HSI denoising methods [18], [31].

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TABLE VII

QUANTITATIVE ASSESSMENT OF EIGENIMAGE-BASED METHODS (WITH DIFFERENT SETTING OF SUBSPACE DIMENSION) APPLIED TO SIMULATED WASHINGTON DC MALL IMAGES CORRUPTED BY SPATIALLY VARIANT GAUSSIAN NOISE

		Noisy	LRTF-DFR (TGRS, 2020) [19]	NG-meet (TPAMI, 2020) [41]	FastHyMix (TNNLS, 2021) [31]	NL-3DCNN (TGRS, 2022) [71]	MAC-Net (TGRS, 2022) [72]	Eigen-CNN (proposed)
$p = 8$								
Case 9	MPSNR (dB)	19.25	31.41	31.82	35.49	35.73	35.13	**36.70**
	MSSIM	0.6120	0.9390	0.9431	0.9785	0.9794	0.9765	**0.9858**
	MFSIM	0.7900	0.9593	0.9636	0.9825	0.9834	0.9810	**0.9872**
$p = 10$								
Case 9	MPSNR (dB)	19.25	30.27	30.09	35.37	35.59	34.77	**36.64**
	MSSIM	0.6120	0.9212	0.9175	0.9779	0.9787	0.9745	**0.9857**
	MFSIM	0.7900	0.9489	0.9495	0.9822	0.9830	0.9795	**0.9871**
$p = 12$								
Case 9	MPSNR (dB)	19.25	29.47	28.92	35.22	35.40	34.42	**36.53**
	MSSIM	0.6120	0.9100	0.8966	0.9773	0.9780	0.9724	**0.9854**
	MFSIM	0.7900	0.9419	0.9378	0.9818	0.9824	0.9778	**0.9869**

Note: Best results are highlighted in bold.
p denotes the setting of subspace dimension used in LRTF-DFR, NG-meet, FastHyMix, NL-3DCNN, MAC-Net, and Eigen-CNN.

TABLE VIII

QUANTITATIVE ASSESSMENT OF EIGENIMAGE-BASED METHODS (WITH DIFFERENT SETTING OF SUBSPACE DIMENSION) APPLIED TO SIMULATED PAVIA UNIVERSITY IMAGES CORRUPTED BY SPATIALLY VARIANT GAUSSIAN NOISE

		Noisy	LRTF-DFR (TGRS, 2020) [19]	NG-meet (TPAMI, 2020) [41]	FastHyMix (TNNLS, 2021) [31]	NL-3DCNN (TGRS, 2022) [71]	MAC-Net (TGRS, 2022) [72]	Eigen-CNN (proposed)
$p = 8$								
Case 10	MPSNR (dB)	18.24	29.67	31.33	35.97	36.28	35.32	**37.43**
	MSSIM	0.2926	0.7834	0.8407	0.9570	0.9589	0.9404	**0.9691**
	MFSIM	0.6129	0.9076	0.9325	0.9728	0.9746	0.9678	**0.9798**
$p = 10$								
Case 10	MPSNR (dB)	18.24	28.55	29.12	35.76	36.17	34.78	**37.38**
	MSSIM	0.2926	0.7443	0.7633	0.9556	0.9583	0.9319	**0.9688**
	MFSIM	0.6129	0.8893	0.8988	0.9721	0.9743	0.9637	**0.9796**
$p = 12$								
Case 10	MPSNR (dB)	18.24	27.57	27.70	35.57	36.06	34.32	**37.32**
	MSSIM	0.2926	0.7066	0.7062	0.9543	0.9576	0.9237	**0.9684**
	MFSIM	0.6129	0.8708	0.8721	0.9714	0.9739	0.9595	**0.9794**

Note: Best results are highlighted in bold.
p denotes the setting of subspace dimension used in LRTF-DFR, NG-meet, FastHyMix, NL-3DCNN, MAC-Net, and Eigen-CNN.

When compared with other traditional machine learning-based methods (namely, NonRLRS, KBR, LRTF-DFR, NG-meet, and FastHyMix) in Tables II and III, Eigen-CNN still performs the best. The reasons for this include: Eigen-CNN effectively utilizes the clean texture features learned from other clean images in the ICVL dataset when training. In addition, the concurrent use of high-quality first eigenimage and eigennoise level map as input data effectively enhances the network's ability to express spatial image information and remove noise. These factors together contribute to the superior performance of Eigen-CNN in denoising tasks, surpassing other methods.

Basically, the eigenimage filtering methods (namely, LRTF-DFR, FastHyMix, NG-meet, and Eigen-CNN) perform better than the sparse and low-rank representation-based methods (namely, NonRLRS and KBR). Among the eigenimage filtering methods, their key difference lies in the way of removing noise remaining in eigenimages. LRTF-DFR minimizes the spatial difference of eigenimages, NG-meet enforces the low-rankness of nonlocal patches extracted from eigenimages, FastHyMix exploits the existing regularizer from a state-of-the-art denoiser, and our Eigen-CNN trains an eigenimage denoising network embedded with a high-quality guide image and an eigennoise level map. Tables II and III show promising results of using neural networks to learn the inherent data structure of eigenimages instead of designing elaborate hand-crafted regularizations for eigenimage denoising.

Considering that LRTF-DFR, NG-meet, FastHyMix, NL-3DCNN, MAC-Net, and Eigen-CNN have the same parameter, i.e., the dimension of subspace, p, and its optimal setting for each method may be different, we set $p \in \{8, 10, 12\}$ and show all results in Tables V and VI for Washington DC Mall and

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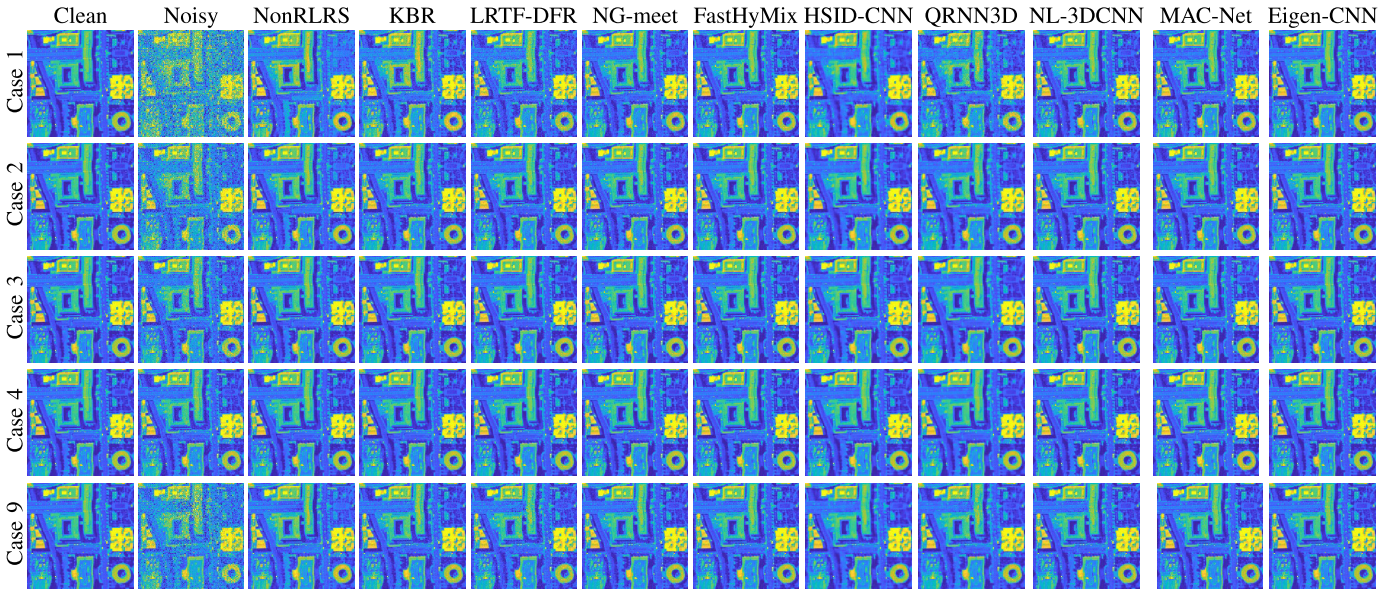


Fig. 4. Band 20 within a simulated subregion of the Washington DC Mall data pre- and post-noise reduction in five cases when p is set to 6.

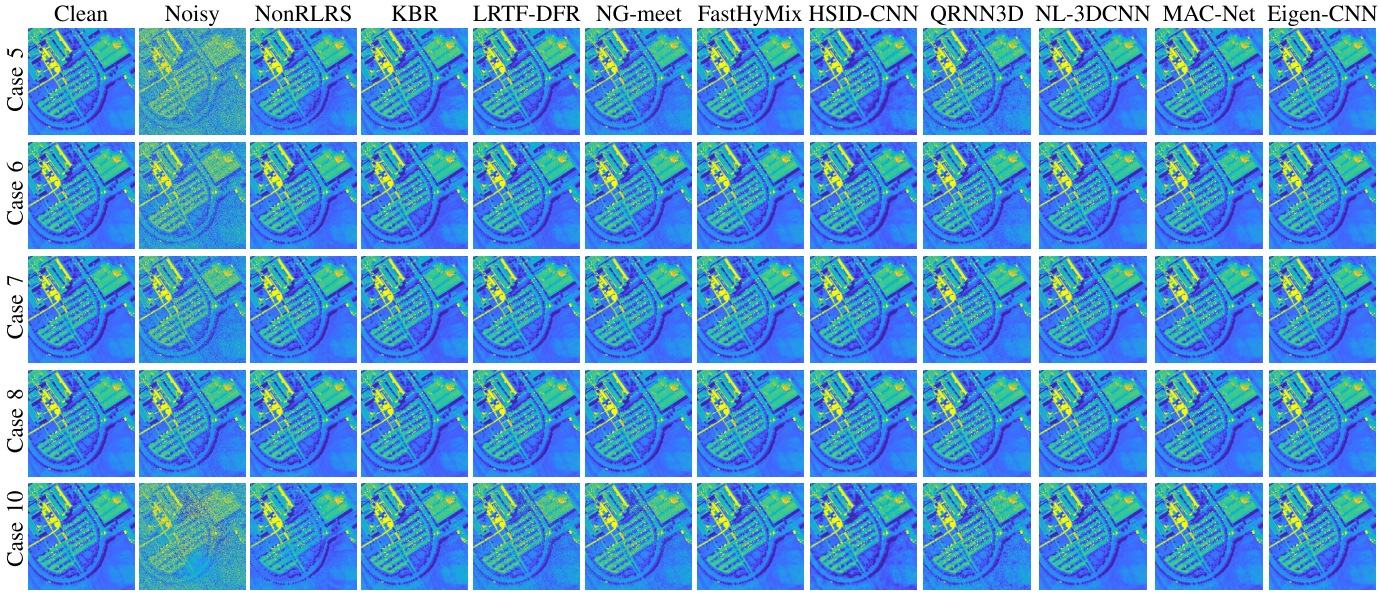


Fig. 5. Band 20 within a simulated subregion of the Pavia University data pre- and post-noise reduction in five cases when p is set to 6.

Pavia University images, respectively (cf. Tables II and III). The increase in parameter p results in the retention of more image signals in the subspace, but it also introduces more noise residuals. One should consider the trade off between signal preservation and noise reduction when adjusting the parameter p. We can observe from Tables V and VI that the denoising performance of RTF-DFR and NG-meet decreases as p increases, whereas FastHyMix, NL-3DCNN, MAC-Net, and Eigen-CNN exhibit relatively stable denoising performance as p increases.

Our Eigen-CNN method has the capability to handle not only Gaussian i.i.d. noise but also spatially variant Gaussian noise, which is characterized by its varying intensities across different spatial locations. This dual-functionality of the Eigen-CNN method is primarily attributed to the incorporation of an eigennoise level map as one of the input data for the network, intended to improve the denoising precision. This strategic input brings forth an additional benefit, enabling our network to process spatially variant Gaussian noise when a nonuniform eigennoise level map is used. To validate this assertion, we generated new noisy images by introducing Gaussian noise with a zero mean and a spatially variant standard deviation to the clean images of the Washington DC Mall and the Pavia University. Fig. 7 depicts a pattern of spatially variant standard deviation of the Gaussian noise added, intended to simulate the noisy Washington DC Mall image in Case 9 and the noisy Pavia University image in Case 10. Table IV presents the denoising results of these two noisy images. Also, we can observe from Tables VII and VIII that the denoising performance of RTF-DFR, NG-meet, and MAC-Net shows a decline with the increasing value of parameter p, while FastHyMix, NL-3DCNN, and

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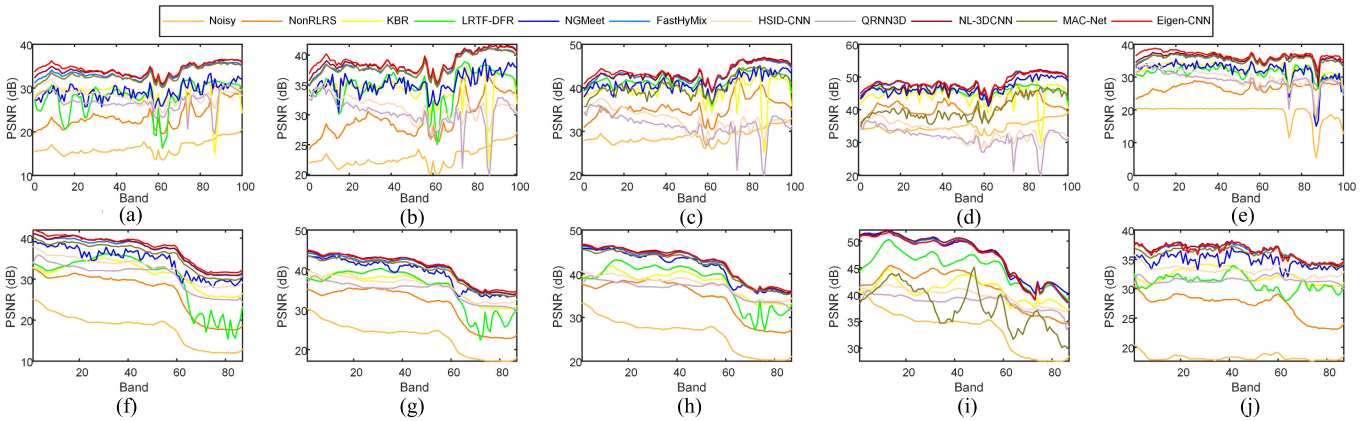


Fig. 6. PSNR metrics across bands for denoised images in Cases 1–10 when p is set to 6. (a) Case 1. (b) Case 2. (c) Case 3. (d) Case 4. (e) Case 5. (f) Case 6. (g) Case 7. (h) Case 8. (i) Case 9. (j) Case 10.

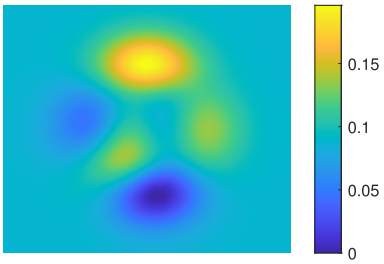


Fig. 7. Spatially variant standard deviation of Gaussian noise added to simulate noisy images in Case 9 and Case 10.

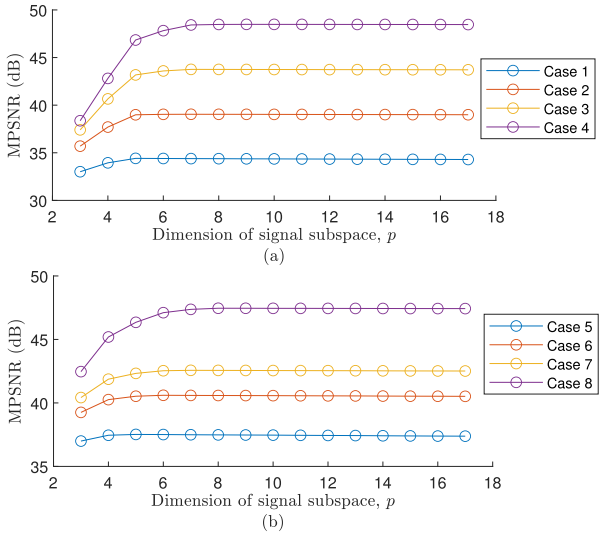


Fig. 8. MPSNR curves generated by Eigen-CNN method in relation to the dimension of subspace estimation for (a) Washington DC Mall dataset and (b) Pavia University dataset.

Eigen-CNN demonstrate a consistent denoising performance even with the increase in p. Notably, the Eigen-CNN method significantly outperformed other methods in terms of precision. This high accuracy of the Eigen-CNN method is attributed to its unique ability to process spatially variant Gaussian noise, a feature that other competitors lack. Therefore, the Eigen-CNN method proves to be a promising tool for handling complex noise scenarios, demonstrating its potential value in the field of geoscience and remote sensing.

C. Impact of Using Guide Image in Eigenimage Denoising Network

To assess the effects of incorporating a guide image in our eigenimage denoising network, we conducted a comparative experiment with simulated datasets. In Table IX, method (e) represents the original version of our denoising network as detailed in Section IV, while method "(c) Eigen-CNN without guide eigenimage" refers to a modified Eigen-CNN that excludes the guide image. Results in Table IX reveal that, in most cases, the Eigen-CNN denoising network with a guide image exhibits superior performance in terms of MPSNR compared with its counterpart that lacks a guide image. This indicates that the utilization of the first eigenimage as a guide image has a beneficial effect on eigenimage denoising.

D. Robustness of Eigen-CNN to Subspace Overestimation

The implementation of the Eigen-CNN framework necessitates only a single parameter, specifically the spectral subspace dimension, denoted by p. This can either be determined empirically or estimated through various signal subspace identification techniques, such as HySime [62] and the VD method [63]. Similar to other advanced eigenimage filtering methodologies (e.g., FastHyDe [18], GLF [27], and FastHyMix [31]), Eigen-CNN's performance remains stable even when subspace overestimation occurs. Fig. 8 demonstrates the consequences of varying subspace dimension configurations on the Eigen-CNN in simulated experimental data. Provided that the subspace dimension is not underestimated, the MPSNR stays relatively constant. An increasing subspace dimension causes an augmentation of noise remaining in corresponding eigenimages; however, this can be effectively mitigated by employing the eigenimage denoising network. As a result, the Eigen-CNN framework exhibits robustness in the presence of subspace dimension overestimation.

E. Ablation Experiments

In our study, we aimed to acquire a deeper understanding of the individual contributions of each element within our

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TABLE IX
PERFORMANCE COMPARISON OF THE PROPOSED EIGEN-CNN FRAMEWORK AND ITS VARIANTS IN TERMS OF MPSNR (DB) INSIMULATED DATASETS

	Case 1	Case 2	Case 3	Case 4	Case 5	Case 6	Case 7	Case 8
Noisy	16.40	22.84	28.86	34.88	17.87	22.98	26.08	33.44
(a) $\widehat{\mathcal{X}} = \mathcal{Y} \times_3 \mathbf{E}^T \times_3 \mathbf{E}$	27.35	33.76	39.87	45.89	28.40	33.49	36.60	44.13
(b) Eigen-CNN with neither guide eigenimage nor eigennoise level map, i.e., $\widehat{\mathbf{Z}}_i = f_\theta(\widetilde{\mathbf{Z}}_i)$	33.75	38.45	42.73	46.10	36.93	40.09	42.00	46.14
(c) Eigen-CNN without guide eigenimage, i.e., $\widehat{\mathbf{Z}}_i = f_\theta(\widetilde{\mathbf{Z}}_i, \widetilde{\mathbf{O}}_i)$	33.94	38.74	43.59	**48.51**	37.23	40.45	42.46	**47.50**
(d) Eigen-CNN without eigennoise level map, i.e., $\widehat{\mathbf{Z}}_i = f_\theta(\widetilde{\mathbf{Z}}_1, \widetilde{\mathbf{Z}}_i)$	34.19	38.54	42.15	44.61	37.22	40.30	42.14	46.03
(e) Eigen-CNN (proposed), i.e., $\widehat{\mathbf{Z}}_i = f_\theta(\widetilde{\mathbf{Z}}_1, \widetilde{\mathbf{Z}}_i, \widetilde{\mathbf{O}}_i)$	**34.38**	**39.04**	**43.75**	48.48	**37.48**	**40.58**	**42.56**	47.46

Note: Best results are highlighted in bold.
Variant (a): We simply project the noisy spectral vectors onto a subspace and do not do denoising on eigenimages, i.e., $\mathcal{Y} \times_3 \mathbf{E}^T \times_3 \mathbf{E}$.

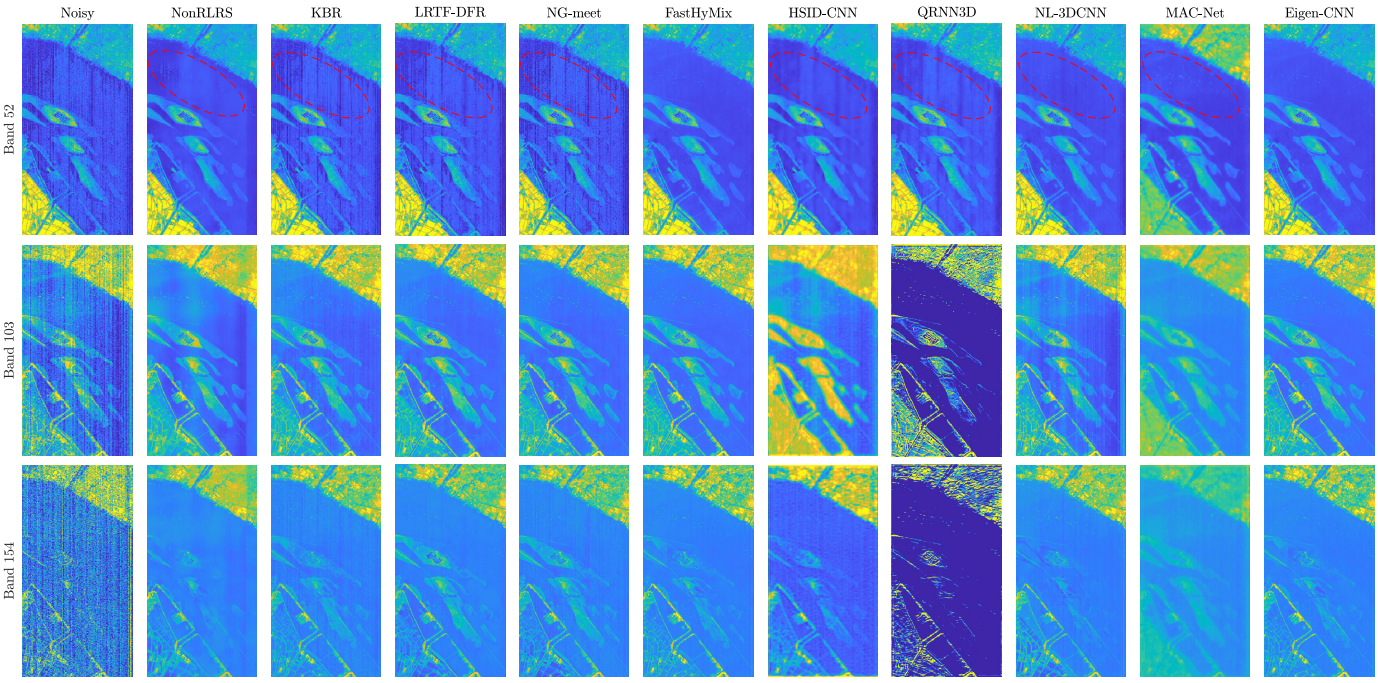


Fig. 9. Denoising performance comparison of NonRLRS (142 s), KBR (1232 s), LRTF-DFR (127 s), NG-meet (40 s), FastHyMix (24 s), HSID-CNN (9 s), QRNN3D (7 s), NL-3DCNN (333 s), MAC-Net (99 s), and Eigen-CNN (12 s) on the real HSI, River image.

Eigen-CNN framework to HSI denoising. To achieve this, we performed ablation experiments on simulated images in Cases 1–8. We evaluated various versions of the Eigen-CNN framework in this process.

The ablation findings are presented in Table IX. Here, the framework "(e) Eigen-CNN" is our proposed method outlined in Section IV and serves as a reference point for evaluating restoration performance. An analysis of variant (b) alongside methods (c)–(e) in Table IX offers insights into the potential PSNR improvements when employing distinct denoising strategies for eigenimages. In particular, examining the differences between variants (b) and (e) reveals that the integration of guide eigenimages and eigennoise-level maps substantially and consistently boosts Eigen-CNN's performance in all cases. Furthermore, the results of variants (e) and (d) suggest that both components (namely, guide eigenimages and eigennoise level maps) are crucial for improving performance.

VI. EXPERIMENTS ON REAL HSIS

We evaluated the denoising performance of our Eigen-CNN on two real HSIs, namely, the River image and the Hyperion Cuprite image, as shown in Fig. 3(c) and (d).

A. River Image

The River image [75], captured by the Earth Observing-1 Hyperion sensor on December 31, 2013, in Jiangsu Province, China, serves as the first real image for our tests. This image comprises 463 × 241 pixels in the spatial dimension and 198 bands in the spectral dimension. As it is a real image, we do not have its ground-truth image for calculating the image denoising accuracy. Therefore, we present the visual effects of denoising by various methods in Fig. 9, with the time taken by each method indicated in the figure captions. The noise in the 52nd, 103rd, and 154th bands is particularly significant.

In the comparative algorithms, we observed that NonRLRS and MAC-Net effectively eliminate noise, but at the cost of losing most of the image's detailed information, resulting in overly smoothed images. The denoising results of KBR, LRTF-DFR, NG-meet, HSID-CNN, QRNN3D, and NL-3DCNN methods still retain noticeable noise in the 52nd band, as indicated by the areas circled in red dashed lines. Also, we observed an issue of incorrect radiance intensity in the denoising results of the 103rd band for both HSID-CNN and QRNN3D methods. FastHyMix and Eigen-CNN performed

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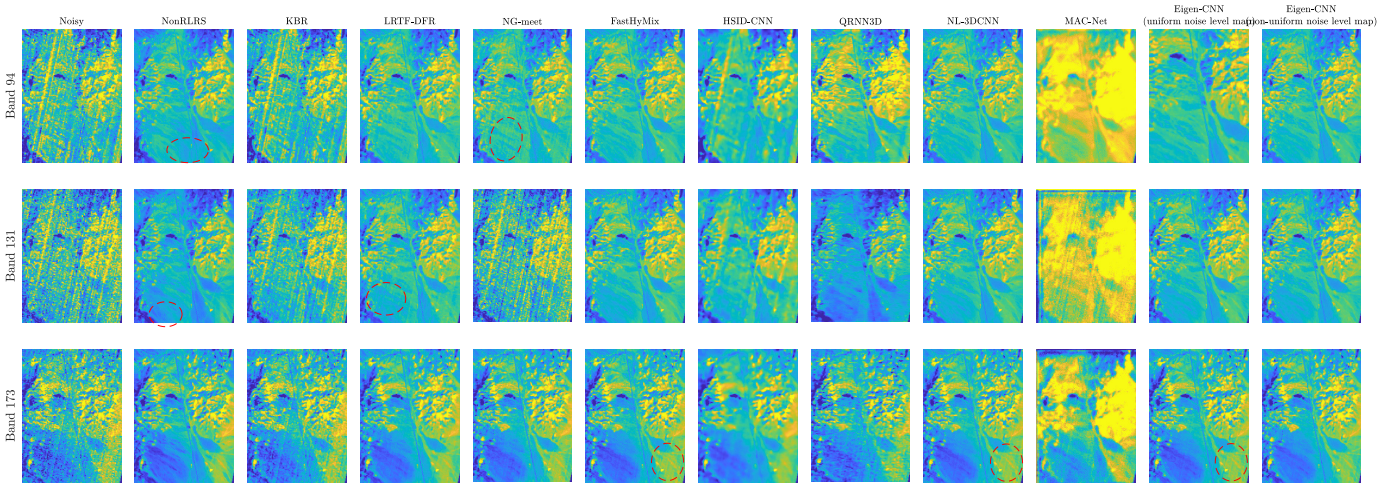


Fig. 10. Denoising performance comparison of NonRLRS (75 s), KBR (431 s), LRTF-DFR (48 s), NG-meet (16 s), FastHyMix (7 s), HSID-CNN (2 s), QRNN3D (2 s), NL-3DCNN (134 s), MAC-Net (48 s), Eigen-CNN with uniform noise level map (4 s), and Eigen-CNN with nonuniform noise level map (4 s) on the real HSI, Hyperion Cuprite image.

the best overall, managing to remove most of the noise while preserving the image details excellently.

B. Hyperion Cuprite Image

The second real image is called Hyperion Cuprite. The Hyperion sensor captured this image in Cuprite, NV, USA, and it contains 242 bands, covering a spectral range of 355–2577 nm with a spectral resolution of 10 nm. The spatial resolution is 30 m. For our experiment, we selected a subimage with dimensions of 240 × 178 pixels, which resulted in 177 remaining bands after removing the water vapor absorption bands.

Since we do not have a ground truth image for the Hyperion Cuprite image, we evaluated the denoising methods visually. Fig. 10 shows three bands that were heavily affected by noise, along with the computation time for each method. From Fig. 10, we can observe that the KBR, NG-meet, and HSID-CNN methods failed to remove noise in the 131st band, and QRNN3D method failed in the 173rd band. MAC-Net obtained incorrect radiance intensity in the denoising results. The other methods were able to effectively reduce most of the noise. However, we have highlighted small residual noise areas using red dashed circles. In the NonRLRS result, the circled area exhibits mosaic-like artifacts, while the LRTF-DFR result shows slight stripe noise. FastHyMix, NL-3DCNN, and Eigen-CNN demonstrate the least residual noise.

In addition, we tested two versions of the Eigen-CNN method. One version assumed spatially invariant noise and used a uniform noise map, while the other version assumed spatially variant noise and used a nonuniform noise map. Both versions of Eigen-CNN achieved good denoising results. However, in the lower right corner of the 173rd band, it can be observed that the result of "Eigen-CNN (uniform noise map)" still has a faint residual stripe noise, whereas "Eigen-CNN (nonuniform noise map)" performs better, demonstrating that the use of an appropriate nonuniform eigennoise level map in Eigen-CNN can effectively remove complex noise and preserve image details in real images.

VII. Conclusion

This article presents Eigen-CNN, a novel approach to HSI denoising that successfully harnesses the low-rank property of spectral dimensions through eigendecomposition and exploits the high spatial correlation of images through a CNN. This method provides a significant advancement over other deep HSI denoising networks that operate in the original image domain. By operating on eigenimages and allowing for a network that is eigenimage-wise, the Eigen-CNN overcomes the persistent challenge of network generalization and can be adeptly applied to new HSIs regardless of the number of bands. Moreover, the Eigen-CNN improves denoising accuracy by using the high-quality first eigenimage as a guiding mechanism for the network. This is a notable improvement over other single eigenimage-based denoisers that either do not fully utilize other eigenimages or lose network flexibility by inputting all eigenimages. In addition, the Eigen-CNN incorporates a eigennoise level map as one of the network inputs, allowing it to effectively manage spatial-variant noise. The Eigen-CNN, with its efficient noise estimation method for eigenimages, has shown promising results in both simulated and real HSIs. These characteristics put Eigen-CNN in a privileged position to be used as an HSI denoiser. Future work will continue to refine and expand upon the Eigen-CNN methodology to further enhance its application and impact on the field of remote sensing.

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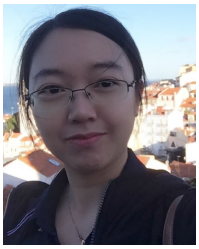
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